

Closure Forces Spherical Geometry: Genuinely Closed Three-Dimensional Systems Are Diffeomorphic to S^3

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Abstract

At the faithful smooth-realization stage of the closure program, a genuinely closed realized three-dimensional system is forced to the spherical case. The theorem proved here is conditional on that stage: if a compact connected simply connected orientable Riemannian 3-manifold is a genuinely frame-complete closed realization, then it is diffeomorphic to S^3 .

The proof separates the closure-program input from the classical differential geometry. The closure input is that admissible frame transport is exactly transport by internal automorphisms of the realized metric structure, hence by elements of $\text{Iso}(M)$, and that no local orthonormal frame is externally distinguished from any other frame. Equivalently, the admissible frame-comparison groupoid is transitive on the orthonormal frame bundle FM . It follows that $\text{Iso}(M)$ acts transitively on FM . Frame transitivity makes the sectional curvature independent of both point and two-plane; the realized metric therefore has constant sectional curvature. The classical space-form classification then leaves only S^3 in the compact simply connected three-dimensional case.

The paper does not rederive the continuum interface from raw comparison data. That interface, including the faithful smooth-realization conditions and the canonical realization, is supplied by the monograph [1].

1 Introduction

This paper proves the spherical rigidity theorem for the manifold-stage realization of the closure program. The statement is not that a sphere is an example of a closed manifold; many compact manifolds have no boundary. The statement is that, after the closure program has reached a faithful smooth Riemannian realization, genuine frame-completeness leaves no three-dimensional compact simply connected realized geometry except the spherical one.

The background closure program is developed in [1]. It begins with primitive comparison data and rectangular completeness, derives the quotient-semantic architecture and its diagonal symmetry action, and then analyzes the conditions under which a faithful smooth Riemannian realization is available. The two-subsystem finitary obstruction theory used by that program is isolated in [2].

The present paper starts only after that interface has been reached. Thus the logical position of the result is:

- (i) primitive comparison data and rectangular completeness give the closed comparison architecture and quotient semantics [1, 2];
- (ii) the faithful smooth-realization interface supplies a compact connected orientable Riemannian 3-manifold (M, g) , with the stated simple-connectivity condition in the case considered here [1];
- (iii) genuine frame-completeness at that manifold stage forces (M, g) to have constant positive sectional curvature, hence $M \cong S^3$.

Only step (iii) is proved in this paper.

What is meant by closure at this stage. Once a metric manifold has been realized, the internal changes of orthonormal frame that preserve the realized comparison structure are the isometries of (M, g) . Any additional frame transport would introduce a rule not recoverable from the realized internal geometry. Conversely, every isometry preserves all coherent comparison reports. The admissible frame-comparison groupoid is therefore the action groupoid

$$FM // \text{Iso}(M).$$

Genuine frame-completeness is the requirement that this internal comparison groupoid leave no local orthonormal frame outside the comparison domain of any other. That condition is exactly transitivity of the $\text{Iso}(M)$ -action on FM .

Classical geometry used. The remaining argument is standard. If the isometry group acts transitively on the orthonormal frame bundle, then all sectional curvatures are equal; this is proved directly below and is part of the standard homogeneous Riemannian geometry background [5, 4]. The classification of complete constant-curvature Riemannian manifolds then identifies the simply connected compact three-dimensional case with S^3 [6].

2 Manifold-Stage Closure

Let (M, g) be a compact connected orientable Riemannian 3-manifold without boundary. Let FM denote its orthonormal frame bundle: a point of FM is an orthonormal basis $e = (e_1, e_2, e_3)$ of some tangent space $T_x M$. Let $\text{Iso}(M)$ denote the isometry group of (M, g) . Each $h \in \text{Iso}(M)$ acts on frames by pushforward,

$$e = (e_1, e_2, e_3) \mapsto h_* e := (h_* e_1, h_* e_2, h_* e_3).$$

Definition 2.1 (Coherent frame report). *A frame report is coherent if it is invariant under the induced $\text{Iso}(M)$ -action. For example, a report $R : FM^k \rightarrow S$ is coherent when*

$$R(h_* e_1, \dots, h_* e_k) = R(e_1, \dots, e_k) \quad \forall h \in \text{Iso}(M).$$

Equivalently, R factors through the orbit projection $FM^k \rightarrow FM^k / \text{Iso}(M)$.

This is the manifold-stage form of quotient semantics: admissible report content is orbit-level content. The same principle is established for the underlying closed comparison framework in [1, 2].

Definition 2.2 (Admissible frame transport). *An admissible frame transport from $e \in FM$ to $f \in FM$ is a morphism in the action groupoid*

$$FM // \text{Iso}(M).$$

Equivalently, it is an isometry $h \in \text{Iso}(M)$ satisfying $h_ e = f$.*

Remark 2.3 (Canonical comparison target). *The action groupoid is not an arbitrary choice of target. At the realized Riemannian stage, the only transformations preserving all metric comparison data are isometries. Adding a frame morphism not induced by an isometry would add transport data that cannot be read from the realized closed system itself. Removing an isometry-induced morphism would discard an internal symmetry already present in the realized structure. Thus $FM // \text{Iso}(M)$ is the canonical admissible frame-comparison groupoid at this stage.*

Definition 2.4 (Frame-complete closed realization). *A manifold-stage closed realization (M, g) is genuinely frame-complete if for every pair of frames $e, f \in FM$ there is an admissible frame transport from e to f .*

Proposition 2.5 (Frame completeness and transitivity). *For a manifold-stage closed realization (M, g) , genuine frame-completeness is equivalent to transitivity of the $\text{Iso}(M)$ -action on FM .*

Proof. By Theorem 2.2, an admissible transport from e to f is exactly an isometry $h \in \text{Iso}(M)$ with $h_* e = f$. Such an isometry exists for every pair $e, f \in FM$ precisely when the $\text{Iso}(M)$ -action on FM is transitive. $\square$

Remark 2.6 (No homogeneous-space shortcut). *The proof does not use the false assertion that a G -equivariant map between homogeneous G -sets exists only when the two points lie in the same orbit. In general, maps $G/H \rightarrow G/K$ are controlled by subgroup containment up to conjugacy. The relevant closed-system statement here is instead direct: the admissible frame morphisms are the morphisms of $FM // \text{Iso}(M)$, and frame-completeness asks whether those morphisms connect every pair of frames.*

3 Curvature Rigidity

Lemma 3.1 (Frame transitivity implies constant curvature). *Let (M, g) be a connected Riemannian manifold. If $\text{Iso}(M)$ acts transitively on the orthonormal frame bundle FM , then (M, g) has constant sectional curvature.*

Proof. Let $\sigma \subset T_x M$ and $\tau \subset T_y M$ be arbitrary two-dimensional tangent planes. Choose orthonormal frames $e = (e_1, \dots, e_n)$ at x and $f = (f_1, \dots, f_n)$ at y such that

$$\sigma = \text{span}\{e_1, e_2\}, \quad \tau = \text{span}\{f_1, f_2\}.$$

By transitivity on FM , there is an isometry $h \in \text{Iso}(M)$ with $h_*e = f$. Isometries preserve the Riemann curvature tensor, hence preserve sectional curvature:

$$K_x(\sigma) = K_y(\tau).$$

Since x, y, σ , and τ were arbitrary, the sectional curvature is independent of both point and two-plane. Thus (M, g) has constant sectional curvature. $\square$

Remark 3.2 (Relation to homogeneous geometry). *Theorem 3.1 is the elementary curvature content behind the standard statement that a Riemannian manifold whose isometry group is transitive on orthonormal frames is a real space form. For the classical background on frame bundles, isometries, and homogeneous Riemannian geometry, see [5, 4, 6].*

4 Main Theorem

Theorem 4.1 (Frame-closure spherical rigidity). *Let (M, g) be a compact connected simply connected orientable Riemannian 3-manifold without boundary. If (M, g) is a genuinely frame-complete closed realization in the sense of Theorem 2.4, then M is diffeomorphic to S^3 .*

Proof. By Theorem 2.5, genuine frame-completeness is equivalent to transitivity of the $\text{Iso}(M)$ action on FM . By Theorem 3.1, (M, g) has constant sectional curvature κ .

Since M is compact, it is geodesically complete. By the space-form classification, a complete connected Riemannian 3-manifold of constant sectional curvature is a quotient of the corresponding simply connected model space:

$$S_\kappa^3, \quad \mathbb{R}^3, \quad \text{or} \quad H_\kappa^3,$$

according as $\kappa > 0$, $\kappa = 0$, or $\kappa < 0$ [6]. Because M is simply connected, it is itself the simply connected model space. The flat and hyperbolic simply connected model spaces, $\mathbb{R}^3$ and H_κ^3 , are noncompact, whereas M is compact. They are therefore excluded. Thus $\kappa > 0$ and M is isometric to a round 3-sphere of radius $1/\sqrt{\kappa}$. In particular, M is diffeomorphic to S^3 . $\square$

Corollary 4.2 (Spherical space-form variant). *If the simple-connectivity hypothesis in Theorem 4.1 is removed, the same argument gives that any compact connected orientable genuinely frame-complete closed realization has constant positive sectional curvature and is therefore a spherical space form S^3/Γ .*

Proof. The proof of Theorem 4.1 gives constant curvature. Compactness excludes the simply connected flat and hyperbolic model spaces at the universal-cover level. The universal cover is therefore S^3 , and M is a quotient S^3/Γ by a finite group of deck transformations acting freely and properly discontinuously. $\square$

5 Logical Status

The theorem is a manifold-stage rigidity result. It should not be read as a derivation of smooth manifolds from raw comparison data inside this short paper. That derivation belongs to the monograph's faithful-realization analysis [1].

The load-bearing closure condition used here is Theorem 2.4. Its content is narrow and explicit: once a faithful Riemannian realization has been obtained, no local orthonormal frame is permitted to carry an external label or distinction that prevents it from being internally compared with any other frame. Since the admissible frame transports at the realized stage are isometries, that condition is exactly frame-bundle transitivity.

Thus the proof does not assume spherical topology. It assumes the manifold-stage closed-realization hypotheses and derives spherical geometry from the frame-level comparison condition. The use of simple connectivity is final and classificatory: it removes nontrivial spherical quotients after the positive constant curvature case has already been forced.

6 Relation to the Poincaré Theorem

Perelman's proof of the Poincaré conjecture answers a topological question: a compact simply connected 3-manifold is homeomorphic to S^3 [8, 9, 10]. Hamilton's Ricci-flow program supplied the analytic flow framework in which that proof is situated [7].

The present result has a different hypothesis and a different role. It assumes a Riemannian manifold-stage closed realization and proves that the closure condition forces constant positive curvature. Simple connectivity is not used to generate the geometry; it is used only to select S^3 from the spherical space forms. Accordingly, this paper is not a replacement for the Poincaré theorem. It is a closure-theoretic rigidity statement whose final classification step is compatible with the classical space-form picture.

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